Erdős–Straus Selectors at the Koide–S^6 Determinant Boundary

Exact incidence maps, primitive rays, terminal homology, and information loss

Technical research continuation

3 September 2026

Exact source basis. The source objects are the active Erdős–Straus project archive of The Clankers and the current canonical TeX/Bib bundle for the Alpöge–Fable/Claude S^6 construction and its integrated audit. The two supplied files have SHA-256 values `d1e371380e9507a4e142dbf56430bc37145bf5ba42edfb76d21e898f0d131fea` and `4a8453f2872dbd93269fc53c321cc19b74553fe0a4fa3f31c79644e28203c92a`, respectively. Every map below displays its domain, codomain, inverse where it exists, and lost data where it does not. No regular S^6 torus fibre is identified with a Koide point: the relevant coefficient matrix is singular.

Abstract

For a prime $p \equiv 1 \pmod 4$, every ordered Erdős–Straus solution with smallest denominator a lies on a shell

$$Ryz = pa(y+z), \qquad 4a = p + R, \qquad R \equiv 3 \pmod 4.$$

The maintained divisor selector rewrites the shell by a divisor $u \mid a^2$ and one of two congruences. We prove that the same selector state is exactly an integral point on the incidence of two determinant loci:

$$\det \begin{pmatrix} R & -u \\ 4 & -T \end{pmatrix} = -p^{\varepsilon}, \qquad \det \begin{pmatrix} u & a \\ a & v \end{pmatrix} = 0, \qquad uv = a^2.$$

The first matrix is unimodularly row-equivalent to the terminal relation matrix of the $(3, 4, \infty)$ S^6 construction; the second is a positive integral rank-one matrix and is exactly the Koide/S^6 determinant-boundary form. A unique factorization

$$u = hr^2, \qquad a = hrs, \qquad v = hs^2, \qquad \gcd(r, s) = 1$$

turns the two divisor channels into the linear congruences $R \mid pr + s$ and $R \mid r + s$ and yields closed denominator formulas. The pair consisting of the rank-one matrix and terminal matrix is a globally injective certificate; explicit inverse formulas recover p, ε, j, u and the unit-fraction denominators. The exterior channel has terminal determinant -1 and the printed integral homology package of S^6; the middle channel has determinant $-p$ and p-torsion in degrees one through four. The selector-to-solution map is also determined exactly: exterior states are faithful, while middle states are paired by the fixed-point-free divisor complement involution. Finally, the full Erdős–Straus conjecture is equivalent to nonemptiness of the displayed two-determinant incidence for every prime $p \equiv 1 \pmod{12}$. The construction does not prove that nonemptiness.

1 The fixed-prime shell

Let $p \equiv 1 \pmod 4$ be prime. Write ES(p) for the assertion that there exist positive integers $x \le y \le z$ with

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

Proposition 1.1 (First-half denominator)**.** *Every solution of* (1) *satisfies*

$$\frac{p}{4} < x \le \frac{p}{2}. \tag{2}$$

Consequently there is a unique integer j with

$$0 \le j < \frac{p-1}{4}, \qquad R_j = 4j+3, \qquad x = a_j = \frac{p+R_j}{4}. \tag{3}$$

Proof. The strict lower bound follows from $1/x < 4/p$. Suppose $x > p/2$. Clearing denominators gives

$$4xyz = p(xy + xz + yz),$$

so $p \mid xyz$. If $p \mid x$, all three denominators are at least p and the sum is at most $3/p$. If $p \mid y$, then $y, z \ge p$ and $1/x + 1/y + 1/z < 2/p + 2/p$. Hence $z = kp$. Substitution gives

$$(4k-1)xy = kp(x+y).$$

Neither x nor y is divisible by p, so $4k - 1 = \ell p$. Because $p \equiv 1 \pmod 4$, $\ell \equiv 3 \pmod 4$ and $\ell \ge 3$. After cancelling p,

$$x \le \frac{2k}{\ell} = \frac{p}{2} + \frac{1}{2\ell} < \frac{p+1}{2},$$

contradicting the integral inequality $x > p/2$. Formula (3) is the unique enumeration of the integers in (2). □

Fix j and abbreviate

$$R = R_j = 4j+3, \qquad a = a_j = \frac{p+R}{4}, \qquad N = pa. \tag{4}$$

Then

$$\gcd(R,p) = \gcd(R,a) = \gcd(p,a) = 1. \tag{5}$$

After fixing $x = a$, equation (1) is

$$Ryz = N(y+z), \tag{6}$$

which is equivalent to

$$(Ry - N)(Rz - N) = N^2. \tag{7}$$

Thus, with

$$d = Ry - N, \qquad e = Rz - N, \tag{8}$$

we have $d, e > 0$, $de = N^2$, and

$$y = \frac{N+d}{R}, \qquad z = \frac{N+e}{R}. \tag{9}$$

Conversely, any positive divisor $d \mid N^2$ satisfying $d \equiv -N \pmod R$ gives (6) through (9); its complementary divisor automatically satisfies the same congruence.

2 The two complete divisor channels

Because $\gcd(p, a) = 1$, the p-adic exponent of d is 0, 1, or 2. The involution $d \leftrightarrow e$ exchanges exponents 0 and 2. Retain the exponent-two representative in that pair and define

$$E_j = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + 1\}, \tag{10}$$

$$M_j = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + p\}. \tag{11}$$

The letter E records the exterior p-adic channel $d = p^2u$; the letter M records the middle channel $d = pu$.

Theorem 2.1 (Complete two-channel selector)**.** *For $p \equiv 1 \pmod 4$ prime,*

$$\mathrm{ES}(p) \iff \bigsqcup_{0 \le j < (p-1)/4} (E_j \sqcup M_j) \neq \varnothing. \tag{12}$$

For a state (ε, j, u), where $\varepsilon = 0$ on E_j and $\varepsilon = 1$ on M_j, put

$$T = \frac{4u + p^\varepsilon}{R_j}, \qquad K = \frac{p^{1-\varepsilon}T + 1}{4}. \tag{13}$$

Then $T, K \in \mathbb{Z}_{>0}$ and the corresponding denominators are

$$(x, y, z) = \left(a_j, \frac{p^\varepsilon a_j K}{u}, pK\right), \tag{14}$$

up to interchange of y, z.

Proof. If $d = p^2u$, then $e = a^2/u$ and

$$R \mid d + N \iff R \mid pu + a \iff R \mid 4u + 1.$$

If $d = pu$, then $e = pa^2/u$ and

$$R \mid d + N \iff R \mid u + a \iff R \mid 4u + p.$$

This proves completeness under the factor involution. In either channel, $RT = 4u + p^\varepsilon$. Since $R \equiv 3 \pmod 4$ and $p^\varepsilon \equiv 1 \pmod 4$, one has $T \equiv 3 \pmod 4$, so K is integral. The identities

$$RK = a + p^{1-\varepsilon}u, \qquad Ta = u + p^\varepsilon K$$

give the divisibility in (14); direct substitution then yields (1). □

3 The terminal determinant

Define

$$\lambda(\ell_0, \ell_1, \ell_2) = 12\ell_0 - 4\ell_1 - 3\ell_2 \tag{15}$$

and the ordered terminal relation matrix

$$\mathsf{R}(\ell_0, \ell_1, \ell_2) = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det \mathsf{R} = -\lambda. \tag{16}$$

For a selector state set

$$\ell_0 = \frac{T + 1}{4}, \qquad \ell_1 = u - jT, \qquad \ell_2 = 1. \tag{17}$$

Theorem 3.1 (Exact selector-to-terminal morphism)**.** *For every prime $p \equiv 1 \pmod 4$ and every selector state,*

$$\lambda(\ell_0, \ell_1, 1) = p^{\varepsilon} \tag{18}$$

and

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1 - 4\ell_0 \end{pmatrix} = \mathsf{R}(\ell_0, \ell_1, 1). \tag{19}$$

The inverse is

$$T = 4\ell_0 - 1, \qquad u = \ell_1 + jT. \tag{20}$$

The crosswalk therefore needs only $p \equiv 1 \pmod 4$, not the stronger restriction $p \equiv 1 \pmod{12}$ used to isolate the hard prime class in the source chapter.

Proof. Substituting (17), $R_j = 4j + 3$, and $R_j T = 4u + p^{\varepsilon}$ gives

$$\lambda(\ell_0, \ell_1, 1) = 3(T + 1) - 4(u - jT) - 3 = (4j + 3)T - 4u = p^{\varepsilon}.$$

Matrix multiplication proves (19); (20) is literal inversion. The parity argument uses only $p^{\varepsilon} \equiv 1 \pmod 4$. The S^6 finite-filling admissibility conditions on this slice are also automatic: $\ell_2 = 1$ is odd, and

$$p^{\varepsilon} = \lambda(\ell_0, \ell_1, 1) \equiv -\ell_1 \pmod 3.$$

Because $p \neq 3$, this forces $3 \nmid \ell_1$ without assuming $p \equiv 1 \pmod 3$. □

Put

$$\mathsf{A}_{p,\varepsilon,j,u} = \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix}. \tag{21}$$

Proposition 3.2 (Exact adjugate factorization)**.** *Every selector state satisfies*

$$\mathsf{A}_{p,\varepsilon,j,u} \begin{pmatrix} T & -u \\ 4 & -R_j \end{pmatrix} = p^{\varepsilon} I_2. \tag{22}$$

Thus

$$\mathsf{A}^{-1}_{p,\varepsilon,j,u} = \frac{1}{p^{\varepsilon}} \begin{pmatrix} T & -u \\ 4 & -R_j \end{pmatrix}. \tag{23}$$

For the primitive rank-one completion below,

$$\Gamma(a, u) \begin{pmatrix} s \\ -r \end{pmatrix} = 0, \qquad \gcd(r, s) = 1. \tag{24}$$

Hence a selector state carries simultaneously an index-p^{ε} nonsingular relation matrix and a singular positive rank-one matrix with a primitive integral kernel vector.

Proof. The four entries in the product on the left of (22) are

$$R_j T - 4u = p^{\varepsilon}, \qquad -R_j u + u R_j = 0, \qquad 4T - 4T = 0, \qquad -4u + T R_j = p^{\varepsilon}.$$

This proves the factorization and inverse. Substitution of $\Gamma = h\binom{r}{s}(r \ s)$ gives $\Gamma(s, -r)^{\mathsf{T}} = 0$, and primitivity is $\gcd(r, s) = 1$. □

The left factor in (19) lies in $\mathrm{SL}_2(\mathbb{Z})$. Hence both matrices present the same cokernel. Since the terminal matrix contains entries 3 and 4, the gcd of all its entries is one, and

$$\mathrm{SNF}(\mathsf{R}) = \mathrm{diag}(1, p^{\varepsilon}), \qquad \operatorname{coker} \mathsf{R} \cong \mathbb{Z}/p^{\varepsilon}\mathbb{Z}. \tag{25}$$

The numerical appearance of the pair $(3, 4)$ is exact: the ES residue $R_j = 4j + 3$ is changed to 3 by subtracting j times the second relation, while the coefficient 4 remains fixed. These are the same two integers as the finite orbifold multiplicities in the terminal S^6 presentation.

4 The rank-one completion of the divisor gate

Let $u \mid a^2$ and set

$$v = \frac{a^2}{u}. \tag{26}$$

Define

$$\Gamma(a,u) = \begin{pmatrix} u & a \\ a & v \end{pmatrix} \in \mathrm{Sym}_2(\mathbb{Z}). \tag{27}$$

Proposition 4.1 (Divisibility as a unique rank-one completion)**.** *For positive integers a, u, the following are equivalent:*
(i) $u \mid a^2$;
(ii) there is a positive integer v for which $\det\left(\begin{smallmatrix} u & a \\ a & v \end{smallmatrix}\right) = 0$.
When these conditions hold, $v = a^2/u$ is unique, $\Gamma(a,u)$ is positive semidefinite of rank one, and the map

$$(a,u) \longmapsto \Gamma(a,u) \tag{28}$$

has inverse $\Gamma \mapsto (\Gamma_{12}, \Gamma_{11})$ on its image.

Proof. The determinant equation is $uv - a^2 = 0$. It has an integral positive solution v exactly when $u \mid a^2$, and then v is forced. The trace is positive and the determinant is zero, so the eigenvalues are $u + v$ and zero. □

Theorem 4.2 (Unique primitive-ray factorization)**.** *Every matrix* (27) *has a unique factorization*

$$u = hr^2, \qquad a = hrs, \qquad v = hs^2, \qquad h, r, s \in \mathbb{Z}_{>0}, \qquad \gcd(r,s) = 1. \tag{29}$$

Equivalently,

$$\Gamma(a,u) = h \begin{pmatrix} r \\ s \end{pmatrix} (r \;\; s), \qquad \mathrm{SNF}(\Gamma) = \mathrm{diag}(h, 0). \tag{30}$$

Proof. Put $g = \gcd(a,u)$, $r = u/g$, and $s = a/g$. Then $\gcd(r,s) = 1$. The divisibility $u \mid a^2$ gives $gr \mid g^2 s^2$, so $r \mid gs^2$. Coprimality gives $r \mid g$; put $h = g/r$. Then $u = hr^2$, $a = hrs$, and $v = a^2/u = hs^2$. Conversely these equations clearly produce (27). Any second primitive factorization has $r/s = u/a$ in lowest terms, so it has the same r, s, and then the same h. The gcd of all four matrix entries is h, while its determinant is zero, which proves the Smith form. □

5 The selector becomes two linear congruences

Theorem 5.1 (Primitive channel equations)**.** *Let* (29) *hold. Then*

$$u \in E_j \iff R_j \mid pr + s, \tag{31}$$

$$u \in M_j \iff R_j \mid r + s. \tag{32}$$

When (31) *holds, put*

$$\kappa_E = \frac{pr + s}{R_j}. \tag{33}$$

Then an ordered denominator triple is

$$(x, y, z) = \big(hrs,\ phr\kappa_E,\ hs\kappa_E\big). \tag{34}$$

When (32) *holds, put*

$$\kappa_M = \frac{r + s}{R_j}. \tag{35}$$

Then an ordered denominator triple is

$$(x, y, z) = \big(hrs,\ phr\kappa_M,\ phs\kappa_M\big). \tag{36}$$

Interchanging the last two entries gives the complementary orientation.

Proof. Modulo $R = R_j$, the identity $4a = p + R$ gives $p \equiv 4hrs$. Since R is coprime to $a = hrs$, each of h, r, s is a unit modulo R. Thus

$$pr + s \equiv s(4hr^2 + 1) \pmod{R},$$

which proves (31). Similarly

$$4u + p \equiv 4hr^2 + 4hrs = 4hr(r + s) \pmod{R},$$

which proves (32).

For (34), a common denominator calculation gives

$$\frac{1}{hrs} + \frac{1}{phr\kappa_E} + \frac{1}{hs\kappa_E} = \frac{p\kappa_E + s + pr}{phrs\kappa_E} = \frac{(p + R)\kappa_E}{phrs\kappa_E} = \frac{4}{p}.$$

The middle-channel calculation is

$$\frac{1}{hrs} + \frac{1}{phr\kappa_M} + \frac{1}{phs\kappa_M} = \frac{p\kappa_M + s + r}{phrs\kappa_M} = \frac{(p + R)\kappa_M}{phrs\kappa_M} = \frac{4}{p}.$$

□

Corollary 5.2 (Channel overlap)**.** *Fix (p, j, u) which lies in at least one of E_j, M_j. It lies in both channels exactly when*

$$R_j \mid p - 1. \tag{37}$$

Proof. The difference between the two primitive congruence numerators is $(p - 1)r$. Since r is a unit modulo R_j, the two congruences are equivalent precisely under (37). The hypothesis that one already holds supplies the other. □

6 Primitive-ray residues and the divisor-complement reflection

Let

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{38}$$

For a rank-one completion $\Gamma = h\binom{r}{s}(r\ s)$ write

$$\rho_R(\Gamma) := sr^{-1} \pmod{R}. \tag{39}$$

This is defined because $\gcd(R, hrs) = 1$.

Proposition 6.1 (The two channels as exact primitive-ray residues)**.** *For the fixed shell (p, R, a),*

$$u \in E_j \iff \rho_R(\Gamma) \equiv -p \pmod{R}, \tag{40}$$

$$u \in M_j \iff \rho_R(\Gamma) \equiv -1 \pmod{R}. \tag{41}$$

Thus the two channels are two exact residue points of the primitive rank-one ray. They coincide exactly when $R \mid p - 1$.

Proof. Theorem 5.1 gives $pr + s \equiv 0 \pmod{R}$ in the exterior channel and $r + s \equiv 0 \pmod{R}$ in the middle channel. Multiplication by the inverse of r proves (40) and (41). Their right sides coincide exactly when $p \equiv 1 \pmod{R}$. □

The divisor-complement map is

$$\sigma_j(u) = v = \frac{a_j^2}{u}, \qquad \Gamma(a_j, v) = J\Gamma(a_j, u)J. \tag{42}$$

Theorem 6.2 (Exact complement action)**.** *For every fixed shell:*

(i) *σ_j preserves M_j, has no fixed point there, and the two states u and v reconstruct the same unordered denominator pair.*

(ii) *If $u \in E_j$, then*

$$v \in E_j \quad \Longleftrightarrow \quad R_j \mid p^2 - 1. \tag{43}$$

(iii) *The diagonal rank-one matrix*

$$\Gamma = a_j \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \tag{44}$$

occurs in the exterior channel exactly when

$$R_j \mid p + 1. \tag{45}$$

In that case, with $\kappa = (p+1)/R_j$, it gives

$$\frac{4}{p} = \frac{1}{a_j} + \frac{1}{pa_j\kappa} + \frac{1}{a_j\kappa}. \tag{46}$$

Proof. Under $u = hr^2, a = hrs, v = hs^2$, the complement interchanges r and s. The middle congruence $R \mid r+s$ is symmetric, and (36) shows that the last two denominators are exchanged. A fixed point would give $u = v = a$, but then the middle condition would imply

$$R \mid 4a + p = 2p + R,$$

hence $R \mid 2$, contradicting $R \geq 3$ odd and $\gcd(R, p) = 1$.

Suppose $u \in E_j$. Modulo R,

$$4u \equiv -1, \qquad 4a \equiv p, \qquad 4v = 4a^2u^{-1} \equiv -p^2.$$

Therefore $4v + 1 \equiv 1 - p^2 \pmod{R}$, proving (43). Finally $u = v = a$ is an exterior state exactly when

$$R \mid 4a + 1 = p + R + 1,$$

which is (45); substitution of $r = s = 1, h = a$ into (34) gives (46). □

Let $\mathbf{r}_K(a, u)$ be the Koide vector defined below in (86). The complement has an exact action already in the original three Koide coordinates.

Proposition 6.3 (Denominator exchange as a Koide reflection)**.** *Define*

$$W_K = \frac{1}{3} \begin{pmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{pmatrix}. \tag{47}$$

Then

$$W_K^{\mathsf{T}} W_K = I, \qquad W_K^2 = I, \qquad \det W_K = -1, \qquad Q_K(W_K r) = Q_K(r), \tag{48}$$

and

$$\mathbf{r}_K(a, v) = W_K \mathbf{r}_K(a, u). \tag{49}$$

In particular, middle-channel denominator exchange is represented exactly by this rational orthogonal reflection on the Koide null cone.

Proof. Conjugation by J sends

$$\left(\frac{u+v}{2}, a, \frac{u-v}{2}\right) \longmapsto \left(\frac{u+v}{2}, a, -\frac{u-v}{2}\right).$$

Thus in Lorentz coordinates the complement is $\operatorname{diag}(1,1,-1)$. Conjugating by the orthogonal matrix $\mathcal{F}$ gives

$$W_K = \mathcal{F}^{-1} \operatorname{diag}(1,1,-1)\mathcal{F},$$

whose direct multiplication is (47). The identities in (48) and (49) follow. □

7 The exact terminal–Koide incidence set

For $\varepsilon \in \{0,1\}$ define $\mathscr{I}_\varepsilon(p)$ to be the set of tuples

$$(j, \ell_0, \ell_1, \Gamma) \tag{50}$$

with

$$0 \le j < \frac{p-1}{4}, \qquad \Gamma = \begin{pmatrix} u & a \\ a & v \end{pmatrix} \in \mathrm{Sym}_2(\mathbb{Z}), \qquad u, a, v > 0, \tag{51}$$

subject to

$$T = 4\ell_0 - 1 > 0, \tag{52}$$

$$a = \frac{p + 4j + 3}{4}, \tag{53}$$

$$u = \ell_1 + jT, \tag{54}$$

$$\lambda(\ell_0, \ell_1, 1) = p^\varepsilon, \tag{55}$$

$$\det \Gamma = uv - a^2 = 0. \tag{56}$$

Theorem 7.1 (Erdős–Straus as a two-determinant incidence problem)**.** *The map*

$$(\varepsilon, j, u) \longmapsto \left(j, \frac{T+1}{4}, u - jT, \begin{pmatrix} u & a_j \\ a_j & a_j^2/u \end{pmatrix}\right) \tag{57}$$

is a bijection from the complete tagged selector to $\mathscr{I}_0(p) \sqcup \mathscr{I}_1(p)$. Its inverse is

$$u = \Gamma_{11}, \qquad \varepsilon = \begin{cases} 0, & \lambda = 1, \\ 1, & \lambda = p, \end{cases} \qquad j \text{ retained.} \tag{58}$$

Consequently

$$\boxed{\mathrm{ES}(p) \iff \mathscr{I}_0(p) \sqcup \mathscr{I}_1(p) \neq \varnothing.} \tag{59}$$

Equations (55) *and* (56) *are respectively the terminal determinant condition*

$$\det \mathsf{R}(\ell_0, \ell_1, 1) = -p^\varepsilon \tag{60}$$

and the Koide/S^6 coefficient-boundary condition

$$\det \Gamma = 0. \tag{61}$$

Proof. The forward map is well defined by Theorems 2.1 and 3.1 and Proposition 4.1. Conversely, (56) gives $v = a^2/u \in \mathbb{Z}_{>0}$, hence $u \mid a^2$. Expanding (55) using (52)–(54) gives

$$(4j+3)T - 4u = p^\varepsilon,$$

so $R_j \mid 4u + p^\varepsilon$. This is exactly the relevant selector condition. Both compositions are the identity entry by entry. □

The theorem identifies the surviving arithmetic difficulty exactly. The terminal determinant equation by itself has many integral solutions. The Erdős–Straus gate is the additional requirement that the shared top row (u, a) admit the positive integral rank-one completion (27).

8 Exact quotient from selector states to denominator triples

Let $\mathscr{S}(p)$ be the set of triples $(x, y, z) \in \mathbb{Z}^3_{>0}$ satisfying (1) and $x \le y \le z$. Put

$$\mathcal{E}(p) = \bigsqcup_j \{j\} \times E_j, \qquad \mathcal{M}(p) = \bigsqcup_j \{j\} \times M_j. \tag{62}$$

By Theorem 6.2,

$$\sigma(j, u) = \left(j, \frac{a_j^2}{u}\right) \tag{63}$$

is a fixed-point-free involution of $\mathcal{M}(p)$.

Theorem 8.1 (Exact solution-set quotient)**.** *Sorting the three denominators in* (34) *and* (36) *induces a bijection*

$$\boxed{\mathcal{E}(p) \sqcup (\mathcal{M}(p)/\langle\sigma\rangle) \xrightarrow{\sim} \mathscr{S}(p).} \tag{64}$$

Consequently

$$|\mathscr{S}(p)| = \sum_j |E_j| + \frac{1}{2}\sum_j |M_j|. \tag{65}$$

Proof. Surjectivity is Theorem 2.1 together with the unique first-denominator index of Proposition 1.1. Given a sorted solution, the value $x = a_j$ recovers j. The unordered residual factors

$$\{d, e\} = \{R_j y - p a_j, R_j z - p a_j\}$$

have p-adic exponent multiset $\{2, 0\}$ in the exterior channel and $\{1, 1\}$ in the middle channel; hence the channel is recovered.

In the exterior channel, the factor of p-adic exponent two uniquely has form $p^2 u$, so it recovers u. Two different exterior states therefore cannot give the same sorted triple. In the middle channel the unordered factors are $\{pu, pv\}$, so the sorted triple recovers exactly the orbit $\{u, v\}$. The involution has no fixed point by Theorem 6.2. This proves both injectivity and (65). □

9 A faithful double certificate

Define the global tagged selector set by taking the disjoint union over all primes $p \equiv 1 \pmod 4$. The double-certificate map is

$$\mathfrak{C}(p, \varepsilon, j, u) = (\Gamma(a_j, u), \mathsf{R}(\ell_0, \ell_1, 1)). \tag{66}$$

Theorem 9.1 (Global injectivity and explicit inverse)**.** *The map* (66) *is injective. For a pair in its image, write*

$$\Gamma = \begin{pmatrix} u & a \\ a & v \end{pmatrix}, \qquad \mathsf{R} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & -T \end{pmatrix}.$$

Then the inverse is

$$j = \frac{u - \ell_1}{T}, \qquad p = 4a - (4j + 3), \qquad \lambda = -\det \mathsf{R}, \qquad \varepsilon = \begin{cases} 0, & \lambda = 1, \\ 1, & \lambda = p. \end{cases} \tag{67}$$

The complementary divisor is $v = \Gamma_{22}$, and Theorem 5.1 then recovers the denominator triple.

Proof. Every formula in (67) follows from a displayed entry of the two matrices: $u = \Gamma_{11}$, $a = \Gamma_{12}$, $T = -\mathsf{R}_{22}$, and $\ell_1 = -\mathsf{R}_{12}$. The source equation $\ell_1 = u - jT$ gives j; $4a = p + 4j + 3$ gives p; the determinant gives the channel. Thus two selector states with the same matrix pair have identical p, ε, j, u. □

The terminal matrix alone is not faithful. At $p = 13$, the exterior states $(j, u) = (0, 2)$ and $(1, 5)$ both give

$$\mathsf{R} = \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix}, \tag{68}$$

but their rank-one matrices are respectively

$$\begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}, \qquad \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}. \tag{69}$$

The same terminal matrix also occurs at $p = 73$, $j = 1$, $u = 5$, where

$$\Gamma = \begin{pmatrix} 5 & 20 \\ 20 & 80 \end{pmatrix}. \tag{70}$$

The matrix pair restores exactly the information lost by the terminal projection.

10 The Koide and S^6 coefficient maps

Define the native S^6 coefficient form

$$G(t, m, b) = \begin{pmatrix} 2t & 2m \\ 2m & b/3 \end{pmatrix} \tag{71}$$

and the native Lorentz dictionary

$$\mathcal{L}(t, m, b) = \left(t + \frac{b}{6},\ 2m,\ t - \frac{b}{6} \right) = (H, X, Y). \tag{72}$$

Its inverse is

$$\mathcal{L}^{-1}(H, X, Y) = \left(\frac{H + Y}{2}, \frac{X}{2}, 3(H - Y) \right). \tag{73}$$

The raw-period dictionary is

$$\mathcal{P}_K(H, X, Y) = \left(H + Y, \frac{X}{\sqrt{6}}, H - Y \right) = (T_{\mathrm{raw}}, M_{\mathrm{raw}}, B_{\mathrm{raw}}). \tag{74}$$

The exact morphism between the two coefficient triples is

$$\mathcal{D}(t, m, b) = \left(2t, \sqrt{\frac{2}{3}}\, m, \frac{b}{3} \right), \qquad \mathcal{D}^{-1}(T, M, B) = \left(\frac{T}{2}, \sqrt{\frac{3}{2}}\, M, 3B \right), \tag{75}$$

and

$$\mathcal{P}_K = \mathcal{D} \circ \mathcal{L}^{-1}. \tag{76}$$

Theorem 10.1 (Selector state to the Koide–S^6 boundary)**.** *For a selector state, put $v = a^2/u$ and define*

$$(t, m, b) = \left(\frac{u}{2}, \frac{a}{2}, 3v \right). \tag{77}$$

Then

$$G(t, m, b) = \Gamma(a, u), \tag{78}$$

$$\mathcal{L}(t,m,b)=\left(\frac{u+v}{2},a,\frac{u-v}{2}\right), \tag{79}$$

and

$$\mathcal{D}(t,m,b)=\left(u,\frac{a}{\sqrt{6}},v\right). \tag{80}$$

Therefore

$$\det G(t,m,b)=\frac{2}{3}(tb-6m^2)=0, \tag{81}$$

$$T_{\mathrm{raw}}B_{\mathrm{raw}}-6M_{\mathrm{raw}}^2=uv-a^2=0. \tag{82}$$

Proof. Each assertion is direct substitution. In particular,

$$G\left(\frac{u}{2},\frac{a}{2},3v\right)=\begin{pmatrix}u & a\\ a & v\end{pmatrix}.$$

Equation (76) follows by applying (75) to (73). □

Let

$$Q_K(r_1,r_2,r_3)=\frac{2}{3}(r_1+r_2+r_3)^2-(r_1^2+r_2^2+r_3^2) \tag{83}$$

and let

$$\mathcal{F}(r_1,r_2,r_3)=\left(\frac{r_1+r_2+r_3}{\sqrt{3}},\frac{r_1-r_3}{\sqrt{2}},\frac{r_1+r_3-2r_2}{\sqrt{6}}\right)=(H,X,Y). \tag{84}$$

Then $\mathcal{F}$ is orthogonal and

$$Q_K=H^2-X^2-Y^2. \tag{85}$$

The exact Koide vector associated with the selector matrix is

$$\mathbf{r}_K(a,u)=\mathcal{F}^{-1}\left(\frac{u+v}{2},a,\frac{u-v}{2}\right), \tag{86}$$

that is,

$$(\mathbf{r}_K)_1=\frac{u+v}{2\sqrt{3}}+\frac{a}{\sqrt{2}}+\frac{u-v}{2\sqrt{6}}, \tag{87}$$

$$(\mathbf{r}_K)_2=\frac{u+v}{2\sqrt{3}}-\frac{u-v}{\sqrt{6}}, \tag{88}$$

$$(\mathbf{r}_K)_3=\frac{u+v}{2\sqrt{3}}-\frac{a}{\sqrt{2}}+\frac{u-v}{2\sqrt{6}}. \tag{89}$$

It satisfies

$$Q_K(\mathbf{r}_K(a,u))=0. \tag{90}$$

Proposition 10.2 (The arithmetic field is necessarily enlarged)**.** *The equation $Q_K(r)=0$ has no nonzero solution in $\mathbb{Q}^3$. Thus every nonzero integral selector matrix gives a Lorentz-rational null point through* (79)*, but its Koide coordinates* (87)–(89) *lie in $\mathbb{Q}(\sqrt{2},\sqrt{3})^3$ and never in $\mathbb{Q}^3\setminus\{0\}$.*

Proof. For rational r_i, clear denominators and put

$$S=r_1+r_2+r_3,\qquad U=r_1-r_2,\qquad V=r_1+r_2-2r_3.$$

A direct invertible rational change of coordinates gives

$$Q_K=\frac{2S^2-3U^2-V^2}{6}. \tag{91}$$

Suppose a primitive nonzero integer triple satisfies $2S^2=3U^2+V^2$. Modulo 3, the equality $V^2\equiv 2S^2$ forces $3\mid S,V$. Dividing by 3 and reducing again modulo 3 forces $3\mid U$. This contradicts primitivity. The field claim follows from the explicit formulas. □

Proposition 10.3 (Exact positive Koide sector)**.** *Put*

$$\theta = \frac{u}{a} = \frac{r}{s} > 0 \tag{92}$$

and

$$\theta_- = \frac{\sqrt{3}-\sqrt{2}}{1+\sqrt{2}}, \qquad \theta_+ = \frac{\sqrt{3}+\sqrt{2}}{1+\sqrt{2}}. \tag{93}$$

The first coordinate in (87) *is always positive, and all three Koide coordinates are positive exactly when*

$$0 < \theta < \theta_- \qquad or \qquad \theta_+ < \theta < 1+\sqrt{2}. \tag{94}$$

Proof. Use $u = a\theta$ and $v = a/\theta$ in (87)–(89). After multiplication by the positive factor $2\sqrt{6}\theta/a$, the second coordinate has sign

$$(\sqrt{2}-2)\theta^2 + (\sqrt{2}+2),$$

which is positive exactly for $\theta < 1+\sqrt{2}$. The third has sign

$$(1+\sqrt{2})\theta^2 - 2\sqrt{3}\theta + (\sqrt{2}-1),$$

whose roots are θ_-, θ_+ and whose leading coefficient is positive. The corresponding expression for the first coordinate has all positive coefficients. Because $\theta \in \mathbb{Q}$ and the three boundary numbers are irrational, equality does not occur. □

Remark 10.4 (Exact image obstruction). The map (77) lands in the determinant-zero boundary of the ambient coefficient space. For every choice of the real parts of τ, μ, β, the canonical real period matrix with these imaginary coefficients satisfies $\det P = 0$. Its four columns therefore do not generate a full lattice in $\mathbb{R}^4$. The map is not a map to a regular complex-torus fibre. The actual radial S^6 cusp instead has $\det P_R = -\delta_R \neq 0$ with $\delta_R \sim R^2$.

11 The full residual shell as a second rank-one matrix

The selector matrix Γ records u, a, v. The complete two-denominator shell records d, N, e in

$$\Gamma_{\mathrm{sh}} = \begin{pmatrix} d & N \\ N & e \end{pmatrix}, \qquad de = N^2. \tag{95}$$

The exact map from the selector matrix is channel dependent:

$$\Gamma^E_{\mathrm{sh}} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \Gamma \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p^2 u & pa \\ pa & v \end{pmatrix}, \tag{96}$$

$$\Gamma^M_{\mathrm{sh}} = p\Gamma = \begin{pmatrix} pu & pa \\ pa & pv \end{pmatrix}. \tag{97}$$

Put

$$\mathbf{L}(\Gamma) = \begin{pmatrix} H \\ X \\ Y \end{pmatrix} = \begin{pmatrix} (u+v)/2 \\ a \\ (u-v)/2 \end{pmatrix}, \qquad Q_L(H,X,Y) = H^2 - X^2 - Y^2. \tag{98}$$

Theorem 11.1 (The two ES channels as exact Lorentz-conformal maps)**.** *The channel maps in* (96)–(97) *induce the following linear maps on the Lorentz coordinates of the rank-one symmetric matrix:*

$$\mathbf{L}(\Gamma^E_{\mathrm{sh}}) = \mathcal{B}_p \mathbf{L}(\Gamma), \qquad \mathcal{B}_p = \begin{pmatrix} \frac{p^2+1}{2} & 0 & \frac{p^2-1}{2} \\ 0 & p & 0 \\ \frac{p^2-1}{2} & 0 & \frac{p^2+1}{2} \end{pmatrix}, \tag{99}$$

$$\mathbf{L}(\Gamma^M_{\mathrm{sh}}) = pI_3 \mathbf{L}(\Gamma). \tag{100}$$

With $J_L = \mathrm{diag}(1,-1,-1)$*,*

$$\mathcal{B}_p^{\mathsf{T}} J_L \mathcal{B}_p = p^2 J_L, \qquad (pI_3)^{\mathsf{T}} J_L (pI_3) = p^2 J_L. \tag{101}$$

Consequently

$$Q_L(\mathbf{L}(\Gamma^E_{\mathrm{sh}})) = Q_L(\mathbf{L}(\Gamma^M_{\mathrm{sh}})) = p^2 Q_L(\mathbf{L}(\Gamma)). \tag{102}$$

On primitive rank-one vectors, the exterior channel sends

$$h\begin{pmatrix} r \\ s \end{pmatrix}(r\ s) \longmapsto h\begin{pmatrix} pr \\ s \end{pmatrix}(pr\ s), \tag{103}$$

whereas the middle channel sends

$$h\begin{pmatrix} r \\ s \end{pmatrix}(r\ s) \longmapsto ph\begin{pmatrix} r \\ s \end{pmatrix}(r\ s). \tag{104}$$

Thus the exterior channel changes the primitive projective ray $[r:s]$ *to* $[pr:s]$*, while the middle channel retains* $[r:s]$ *and multiplies its content by* p*.*

Proof. Under the symmetric-matrix identification

$$\begin{pmatrix} H+Y & X \\ X & H-Y \end{pmatrix},$$

conjugation by $\mathrm{diag}(p,1)$ gives

$$\begin{pmatrix} p^2(H+Y) & pX \\ pX & H-Y \end{pmatrix}.$$

Taking its half-trace, off-diagonal entry, and half diagonal difference gives (99). Multiplication of the symmetric matrix by p gives (100). Direct matrix multiplication proves (101); the quadratic identity and the two primitive-ray formulas follow. □

Corollary 11.2 (Exact boost–dilation factorization)**.** *Let*

$$\mathcal{K}_{HY} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \tag{105}$$

Then

$$\mathcal{K}_{HY}^{\mathsf{T}} J_L + J_L \mathcal{K}_{HY} = 0 \tag{106}$$

and the exterior channel matrix has the exact factorization

$$\mathcal{B}_p = p\exp\big((\log p)\mathcal{K}_{HY}\big) = p\begin{pmatrix}\cosh(\log p) & 0 & \sinh(\log p)\\ 0 & 1 & 0\\ \sinh(\log p) & 0 & \cosh(\log p)\end{pmatrix}. \tag{107}$$

The middle channel is the pure dilation pI_3. If $\theta = s/r$ is the positive primitive slope, then the exterior channel sends

$$\theta \longmapsto \frac{\theta}{p}, \tag{108}$$

while the middle channel retains θ. With

$$\zeta_M(\theta) = -\frac{\theta}{(1+\theta)^2}, \tag{109}$$

one has the exact identity

$$\zeta_E(\theta) = \zeta_M\left(\frac{\theta}{p}\right). \tag{110}$$

Proof. Since

$$\cosh(\log p) = \frac{p+p^{-1}}{2}, \qquad \sinh(\log p) = \frac{p-p^{-1}}{2},$$

multiplication by p gives the three entries of $\mathcal{B}_p$ in (99). Direct multiplication proves (106). The primitive-ray map $[r:s] \mapsto [pr:s]$ gives (108), and

$$-\frac{p\theta}{(p+\theta)^2} = -\frac{\theta/p}{(1+\theta/p)^2}$$

proves (110). □

Both matrices have determinant zero. Their Sarnak coordinates are

$$T_S = \frac{d+e}{2}, \qquad X_S = N, \qquad Y_S = 0, \qquad W_S = \frac{d-e}{2}, \tag{111}$$

and satisfy

$$X_S^2 + W_S^2 - T_S^2 = N^2 - de = 0. \tag{112}$$

Denominator exchange is $W_S \mapsto -W_S$.

Define the exact quarter coordinate

$$\zeta = -\frac{yz}{(y+z)^2}. \tag{113}$$

Then

$$\zeta = -\frac{1}{4} + \frac{1}{4}\left(\frac{W_S}{T_S + X_S}\right)^2. \tag{114}$$

Using Theorem 5.1, this becomes

$$\zeta_E = -\frac{prs}{(pr+s)^2}, \tag{115}$$

$$\zeta_M = -\frac{rs}{(r+s)^2}. \tag{116}$$

Proposition 11.3 (The balanced quarter point is excluded)**.** *No Erdős–Straus shell in these channels has $\zeta = -1/4$.*

Proof. Equality in (114) is equivalent to $d = e$, or $y = z$. In the exterior channel this gives $pr = s$. Primitivity forces $r = 1, s = p$, but then $R \mid pr + s = 2p$, impossible because $R \geq 3$ is odd and $\gcd(R, p) = 1$. In the middle channel equality gives $r = s$, so primitivity forces $r = s = 1$, but then $R \mid r + s = 2$, again impossible. □

The null cone (112) is the $p = 1$ member of the Kocik–Sarnak four-circle family used in the Erdős–Straus archive. The Kocik charged-lepton specialization is the different four-circle member $p = 2/3$, whose equation has coefficient $5/3$. The rigorous Koide relation in this note is instead the explicit three-dimensional Lorentz isometry (84)–(85). These two statements are connected by the displayed maps, not by identifying 1 with $2/3$.

12 The terminal topological dichotomy

The source terminal group is

$$\mathcal{G}(\ell) = \langle c, x, y \mid c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \rangle. \tag{117}$$

Eliminating y makes the presentation abelian with relation matrix (16). Therefore

$$\mathcal{G}(\ell) \cong \mathbb{Z}/p^{\varepsilon}\mathbb{Z}. \tag{118}$$

The source all-degree terminal package gives, for $\lambda = p^{\varepsilon}$,

$$(H_0, \ldots, H_6) \cong (\mathbb{Z}, \mathbb{Z}/p^{\varepsilon}, \mathbb{Z}/p^{\varepsilon}, \mathbb{Z}/p^{\varepsilon}, \mathbb{Z}/p^{\varepsilon}, 0, \mathbb{Z})\,. \tag{119}$$

Here $\mathbb{Z}/1 = 0$. Consequently:

Channel	λ	terminal group	terminal homology package
E	1	0	$H_*(S^6; \mathbb{Z})$
M	p	$\mathbb{Z}/p$	$\mathbb{Z}/p$ in degrees $1, 2, 3, 4$

Thus an E-channel Erdős–Straus witness lands on the homology-sphere terminal slice of the S^6 algebra. An M-channel witness lands on a p-torsion terminal slice. A state in both channels supplies two different terminal matrices over the same rank-one Koide boundary matrix.

13 Worked exact examples

13.1 $p = 5$: extension beyond $p \equiv 1 \pmod{12}$

Here $j = 0$, $R = 3$, $a = 2$, and $u = 2$ lies in E_0. One has

$$\Gamma = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \qquad T = 3, \qquad (\ell_0, \ell_1, \ell_2) = (1, 2, 1),$$

$$\mathsf{R} = \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix}, \qquad \det \mathsf{R} = -1,$$

and

$$\frac{4}{5} = \frac{1}{2} + \frac{1}{20} + \frac{1}{4}.$$

This proves by an explicit state that the terminal crosswalk itself applies to the residue class $p \equiv 5 \pmod{12}$.

13.2 $p = 13$: one rank-one ray, two channels

For $j = 0$, $R = 3$, $a = 4$, $u = 2$, one has

$$\Gamma = \begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 2 \end{pmatrix}(1\ 2).$$

The exterior state gives

$$T_E = 3, \quad \mathsf{R}_E = \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix}, \quad \frac{4}{13} = \frac{1}{4} + \frac{1}{130} + \frac{1}{20}.$$

The same (p, j, u) also satisfies the middle congruence because $3 \mid 12$:

$$T_M = 7, \quad \mathsf{R}_M = \begin{pmatrix} 3 & -2 \\ 4 & -7 \end{pmatrix}, \quad \frac{4}{13} = \frac{1}{4} + \frac{1}{26} + \frac{1}{52}.$$

Thus the same Koide/S^6 boundary ray supports both the sphere-type terminal package and the 13-torsion terminal package.

For $j = 1$, $R = 7$, $a = 5$, $u = 5$, the exterior state has

$$\Gamma = \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}, \qquad \mathsf{R} = \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix}, \qquad \frac{4}{13} = \frac{1}{5} + \frac{1}{130} + \frac{1}{10}.$$

This is the same terminal matrix as the preceding exterior state but a different rank-one matrix; Theorem 9.1 distinguishes them.

14 The full Erdős–Straus conjecture

Let ES denote the assertion that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{120}$$

has a positive integral solution for every integer $n \geq 2$.

Theorem 14.1 (Global reduction to the two-determinant incidence)**.** *The full conjecture is equivalent to*

$$\boxed{\text{ES} \quad \Longleftrightarrow \quad \forall p \in \mathbb{P},\ p \equiv 1 \pmod{12} :\ \mathscr{I}_0(p) \sqcup \mathscr{I}_1(p) \neq \varnothing.} \tag{121}$$

Equivalently,

$$\neg\text{ES} \quad \Longleftrightarrow \quad \exists p \in \mathbb{P},\ p \equiv 1 \pmod{12} :\ \mathscr{I}_0(p) = \mathscr{I}_1(p) = \varnothing. \tag{122}$$

The sphere-type subproblem

$$\forall p \equiv 1 \pmod{12}, \quad \mathscr{I}_0(p) \neq \varnothing \tag{123}$$

is sufficient for ES*; it is not asserted to be necessary.*

Proof. If (120) holds for a divisor $d \mid n$, multiplication of all three denominators by n/d gives a solution for n. It therefore suffices, and is necessary, to treat primes.

For $p = 2$,

$$\frac{4}{2} = \frac{1}{1} + \frac{1}{2} + \frac{1}{2}.$$

For a prime $p \equiv 3 \pmod 4$, put

$$a = \frac{p+1}{4}, \qquad N = pa, \qquad b = N+1, \qquad c = N(N+1).$$

Then

$$\frac{4}{p} - \frac{1}{a} = \frac{1}{N} = \frac{1}{N+1} + \frac{1}{N(N+1)}.$$

For a prime $p \equiv 5 \pmod{12}$, put

$$a = \frac{p+3}{4}, \qquad b = \frac{(p+1)(p+3)}{12}, \qquad c = pb.$$

Then

$$\frac{4}{p} - \frac{1}{a} = \frac{3}{pa} = \frac{1}{b} + \frac{1}{pb}.$$

Every prime greater than three is congruent to $1, 5, 7$, or 11 modulo 12; the last two classes are $3 \pmod 4$. Hence only primes $p \equiv 1 \pmod{12}$ remain. For each of those primes, Theorem 7.1 is exactly the asserted incidence criterion. Taking the logical negation gives (122), and retaining only the $\varepsilon = 0$ alternative gives (123). □

Corollary 14.2 (Global primitive Koide-ray formulation)**.** *The full Erdős–Straus conjecture is equivalent to the following statement: for every prime $p \equiv 1 \pmod{12}$, there exist $h, r, s \in \mathbb{Z}_{>0}$ such that*

$$\gcd(r, s) = 1, \qquad \frac{p}{4} < hrs < \frac{p}{2}, \qquad R := 4hrs - p, \tag{124}$$

and at least one of the two exact ray conditions holds:

$$R \mid pr + s \qquad \text{or} \qquad R \mid r + s. \tag{125}$$

A hard prime $p \equiv 1 \pmod{12}$ is a counterexample exactly when every triple satisfying (124) satisfies

$$R \nmid pr + s \qquad \text{and} \qquad R \nmid r + s. \tag{126}$$

Proof. Given an incidence point, Theorem 4.2 writes $u = hr^2, a = hrs, v = hs^2$. Proposition 1.1 gives the strict interval in (124), and Theorem 5.1 gives (125). Conversely, a triple satisfying (124) has

$$0 < R < p, \qquad R \equiv 3 \pmod 4, \qquad j = \frac{R-3}{4} \in \mathbb{Z}, \qquad a = hrs = \frac{p+R}{4}.$$

Set $u = hr^2$ and $v = hs^2$. Then $uv = a^2$, and either divisibility in (125) is, by Theorem 5.1, exactly one of the two selector channels. The global equivalence follows from Theorem 14.1; its logical negation is (126). □

Corollary 14.3 (Divisor-free hard-prime formulation)**.** *The full Erdős–Straus conjecture is equivalent to the following statement: for every prime $p \equiv 1 \pmod{12}$ there exist positive coprime integers r, s and an integer R such that*

$$0 < R < p, \qquad R \equiv 3 \pmod 4, \qquad 4rs \mid p + R, \tag{127}$$

and

$$R \mid pr + s \qquad \text{or} \qquad R \mid r + s. \tag{128}$$

For such data the content and first denominator are uniquely

$$h = \frac{p+R}{4rs}, \qquad a = hrs = \frac{p+R}{4}. \tag{129}$$

Proof. The data of Corollary 14.2 give $R = 4hrs - p$, hence $p + R = 4hrs$ and therefore $4rs \mid p + R$. Conversely, (127) defines the positive integer h in (129); then $R = 4hrs - p$ and $p/4 < hrs < p/2$. The two divisibility conditions are unchanged, so Corollary 14.2 applies in both directions. □

Proposition 14.4 (Exact middle-channel shell bound)**.** *Suppose a hard-prime selector lies in the middle channel and write*

$$r + s = \kappa R, \qquad \kappa \in \mathbb{Z}_{>0}. \tag{130}$$

Then

$$(4\kappa - 1)R \le p + 4. \tag{131}$$

In particular

$$R \le \frac{p+4}{3}, \tag{132}$$

and for every fixed $K \ge 1$, a middle state with $\kappa \ge K$ satisfies

$$R \le \frac{p+4}{4K-1}. \tag{133}$$

Proof. Using $p = 4hrs - R$ and $r + s = \kappa R$, one has the exact identity

$$p + 4 - (4\kappa - 1)R = 4(hrs - r - s + 1) = 4\big((h-1)rs + (r-1)(s-1)\big). \tag{134}$$

Both terms on the last side are nonnegative, proving (131). The two displayed consequences follow from $\kappa \ge 1$ and $\kappa \ge K$, respectively. Equality in (131) holds exactly when $h = 1$ and $(r-1)(s-1) = 0$. □

15 Exact conclusion for Erdős–Straus

For every prime $p \equiv 1 \pmod 4$, the complete result can be written as the following equality of existence statements:

$$\begin{aligned} \mathrm{ES}(p) \quad \Longleftrightarrow \quad & \exists\, \varepsilon \in \{0,1\},\ 0 \le j < \frac{p-1}{4},\ \ell_0, \ell_1, u, a, v \in \mathbb{Z}, \\ & T = 4\ell_0 - 1 > 0, \quad a = \frac{p + 4j + 3}{4}, \quad u = \ell_1 + jT > 0, \quad v > 0, \\ & 12\ell_0 - 4\ell_1 - 3 = p^{\varepsilon}, \qquad uv = a^2. \end{aligned} \tag{135}$$

The two equations on the last line are exactly

$$\det \mathsf{R}(\ell_0, \ell_1, 1) = -p^{\varepsilon}, \qquad \det \Gamma(a, u) = 0.$$

The first is the terminal determinant of the $(3, 4, \infty)$ sphere construction. The second is the positive rank-one Koide/S^6 coefficient boundary. The coupling equations on the preceding line preserve the prime, first-denominator index, orientation, and divisor information.

The sphere construction therefore supplies an exact terminal-topological interpretation of each selector state and the Koide coefficient form supplies an exact rank-one completion of its divisor gate. What remains in the Erdős–Straus conjecture is precisely the assertion that the integral incidence set in (135) is nonempty for every relevant prime. The S^6 terminal algebra alone does not force $uv = a^2$; the Koide determinant boundary alone does not force $12\ell_0 - 4\ell_1 - 3 = p^{\varepsilon}$. Their coupled integral incidence is the complete selector.

Finally, the positive defect-one Koide/Vieta shell studied elsewhere in the source has

$$\Delta = 3Q_K = -1, \qquad Q_K = -\frac{1}{3},$$

whereas every Erdős–Straus selector and shell matrix constructed here has $Q_K = 0$. An invertible isometry of the quadratic space preserves the value of Q_K, so the defect-one shell is not the Erdős–Straus null shell. The proved linear counting theorem for that defect-one hyperboloid is therefore a separate arithmetic result, not a count of Erdős–Straus witnesses.

Reproducibility

The accompanying exact-integer program verifies every displayed map, congruence, inverse, determinant, denominator identity, quarter-coordinate formula, channel overlap, and double-certificate reconstruction for every prime $p \equiv 1 \pmod 4$ through 2000. It checks 147 primes and 6548 tagged selector states. They give exactly 4963 sorted denominator triples, in agreement with (65); all 6548 double certificates are distinct, while 391 terminal-matrix collision classes demonstrate the information lost by the terminal projection alone. The bounded computation is a verification receipt, not a proof of universal selector nonemptiness.

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